

PAPER_442 — Horsehead Nebula (Barnard 33): Per-System MUGE with Growing Erosion $E(t) = 5 \text{ Myr}$

Source: grok_share_68eb34022.txt — Document 15: “Master Universal Gravity Equation_Horsehead Nebula Reloaded Evolution_03May2025.docx” (lines 4487–4820) **Session:** 119 **CP4 Class:** HorseheadNebulaPerSystemMUGE_GrowingErosion5Myr_Calculator (#97)

1. Overview

PAPER_442 delivers the **complete per-system MUGE** for the Horsehead Nebula (Barnard 33, IC 434 pillar), a dense molecular cloud dark nebula silhouetted against ionized hydrogen emission in Orion, located at $d \approx 400 \text{ pc}$. Mass $M = 1000 M_\odot$, physical height $r = 2.5 \text{ ly} = 2.365 \times 10^{16} \text{ m}$, $z \approx 0$ (local MW).

Novel claim (Q1): First UQFF MUGE for the Horsehead featuring a **growing photoevaporative–radiative erosion function** $E(t) = E_0(1 - e^{-t/\tau_{\text{erosion}}})$ with $E_0 = 0.1$ and $\tau_{\text{erosion}} = 5 \text{ Myr} = 1.578 \times 10^{14} \text{ s}$. This contrasts with PAPER_435 (Pillars of Creation, DECAYING form $E_0 e^{-t/\tau}$) and shares the GROWING form from PAPER_440 (Bubble Nebula, $\tau = 4 \text{ Myr}$). The Horsehead uses $\tau = 5 \text{ Myr}$ because its UV driver (σ Orionis) delivers a softer radiation field than Bubble’s BD+60°2522 OB star, requiring a longer build-up to maximum erosion rate.

2. System Parameters

Parameter	Symbol	Value
Mass	M	$1000 M_\odot = 1.989 \times 10^{33} \text{ kg}$
Scale height	r	$2.5 \text{ ly} = 2.365 \times 10^{16} \text{ m}$
Redshift	z	≈ 0 (local)
H_0		$2.184 \times 10^{-18} \text{ s}^{-1}$
Magnetic field	B	10^{-6} T
Erosion factor	E_0	0.1
Erosion timescale	τ_{erosion}	$5 \text{ Myr} = 1.578 \times 10^{14} \text{ s}$
Wind density	ρ_w	10^{-21} kg/m^3
Wind velocity	v_w	$2 \times 10^6 \text{ m/s}$
Fluid density	ρ_f	10^{-21} kg/m^3

3. Time-Dependent Function

Growing erosion ($E = 0$ at birth, builds to E as UV field establishes ionization front):

$$E(t) = 0.1 \left(1 - e^{-t/\tau_{\text{erosion}}} \right)$$

At $t = 0$: $E(0) = 0$ — no erosion, cloud just formed

At $t = 5 \text{ Myr}$ ($= \tau$): $E = 0.1(1 - e^{-1}) = 0.1 \times 0.6321 = 0.0632$ — 63% of maximum

At $t = 20 \text{ Myr}$ ($\approx 4\tau$): $E \approx 0.0982$ — 98.2% of maximum

At $t \rightarrow \infty$: $E \rightarrow 0.1$

Contrast with Bubble Nebula (PAPER_440): Same form, but $\tau = 4 \text{ Myr}$ vs 5 Myr — 25% longer build-up for the Horsehead due to weaker UV flux from $\sigma \text{ Orionis}$ ($T_{\text{eff}} \approx 35,000 \text{ K}$) versus Bubble's BD+60°2522 ($T_{\text{eff}} \approx 40,000 \text{ K}$).

Contrast with Pillars (PAPER_435): DECAYING $E_0 e^{-t/\tau}$ — Pillars start at maximum erosion and decline (Tremblin et al. 2012 “pillars formed by retreating ionization front”). Horsehead uses GROWING — the ionization front of IC 434 is slowly advancing TOWard the dark cloud from $\sigma \text{ Ori}$.

4. Complete 10-Term MUGE

$$g_{\text{HH}}(r, t) = T_1(1 + E) + T_2(1 + E) + T_3 + T_4 + T_5 + T_6 + T_7 + T_8 + T_9 + T_{10}$$

T1 — Newtonian + H t + B + erosion:

$$T_1 = \frac{GM}{r^2}(1 + H_0 t)(1 - B/B_{\text{crit}})(1 + E(t))$$

$$\frac{GM}{r^2} = \frac{6.674 \times 10^{-11} \times 1.989 \times 10^{33}}{(2.365 \times 10^{16})^2} = \frac{1.327 \times 10^{23}}{5.593 \times 10^{32}} \approx 2.37 \times 10^{-10} \text{ m/s}^2$$

$$T_1(t = 0) = 2.37 \times 10^{-10} \times 1.0 \times (1 - 2.27 \times 10^{-20}) \times 1.0 \approx 2.37 \times 10^{-10} \text{ m/s}^2$$

$$T_1(t = 5 \text{ Myr}) = 2.37 \times 10^{-10} \times 1.063 \approx 2.52 \times 10^{-10} \text{ m/s}^2$$

T2 — UQFF Ug channels:

$$T_2 = 2 \times \frac{GM}{r^2} \times f_{\text{TRZ}} \times (1 + E(t)) \approx 2 \times 2.37 \times 10^{-10} \times 1.1 \times 1.063 \approx 5.53 \times 10^{-10} \text{ m/s}^2 \text{ at } t = 5 \text{ Myr}$$

T3 — Λ dark energy:

$$T_3 = \frac{\Lambda c^2}{3} r = \frac{1.11 \times 10^{-52} \times 9 \times 10^{16}}{3} \times 2.365 \times 10^{16} \approx 7.9 \times 10^{-17} \text{ m/s}^2 \quad [\text{negligible}]$$

T9 — Wind:

$$T_9 = \frac{\rho_w v_w^2}{\rho_f \cdot r} = \frac{10^{-21} \times 4 \times 10^{12}}{10^{-21} \times 2.365 \times 10^{16}} = \frac{4 \times 10^{12}}{2.365 \times 10^{16}} \approx 1.69 \times 10^{-4} \text{ m/s}^2$$

5. Canonical Numerical Result

At $t = 5 \text{ Myr}$ (one erosion timescale):

Term	Value (m/s ²)	Fraction
T_9 Wind/Radiation	1.69×10^{-4}	99.99%
T_2 UQFF $U_g \times (1+E)$	5.53×10^{-10}	0.003%
T_1 Newtonian $\times (1+E)$	2.52×10^{-10}	0.001%
T_3 Λ	$\lesssim 10^{-16}$	negligible

$$\boxed{g_{\text{HH}}(t = 5 \text{ Myr}) \approx 1.69 \times 10^{-4} \text{ m/s}^2} \quad [\text{wind/radiation erosion fully dominant}]$$

Growing erosion contribution at t=5 Myr: $E = 0.0632 \Rightarrow T_1, T_2$ increase by 6.3% — magnitude:

$$\Delta g_E = 0.063 \times (T_1 + T_2) \approx 0.063 \times 7.9 \times 10^{-10} \approx 5 \times 10^{-11} \text{ m/s}^2 \quad [\text{detectable in molecular line kinematics}]$$

6. Uniqueness vs Prior Papers

Prior Paper	Overlap	New in PAPER_442
PAPER_435 (Pillars)	Same $E(t)$ form	GROWING vs DECAYING — opposite physics
PAPER_440 (Bubble Nebula)	Same GROWING $E(t)$ form	=5 Myr vs 4 Myr, B is 10^{-6} not 10^{-5}
None	Small dark nebula pillar case	First Barnard 33 complete MUGE
None	Wind absolute dominance ratio	T9/T2 3×10 — highest ratio in per-system series

7. Comparison to Standard Model

Standard photodissociation region (PDR) models (Hollenbach & Tielens 1999) treat Horsehead erosion as a UV-driven mass-loss rate $\dot{M} \sim 10^{-6} M_\odot/\text{yr}$, implying complete photoevaporation in $\sim 10^9$ yr. The UQFF growing- $E(t)$ formulation reformulates this as a multiplicative enhancement to T_1 and T_2 : the Horsehead’s self-gravity is progressively overridden as UV penetration depth grows. This unification of gravitational suppression and radiation erosion is absent in SM PDR models, which treat gravity as a background effect only.

8. Testable Predictions

Q5 Prediction 1: $\tau_{\text{erosion}} = 5$ Myr predicts that the ionization front of IC 434 is currently at $\sim 40\%$ of its maximum advance speed toward Barnard 33 (given estimated cloud age of ~ 2 Myr $\ll \tau$). UQFF predicts an observed $\text{C}^{18}\text{O } J = 1 \rightarrow 0$ line width increase of $\sim 6\%$ from the base of the pillar to the head — testable with IRAM-30m spectral mapping.

Q5 Prediction 2: At $t = \tau_{\text{erosion}} = 5$ Myr from formation, $E = 0.063 \Rightarrow B$ field-corrected gravity is 6.3% stronger than standard Newtonian. This 6.3% self-gravity enhancement maintains the pillar top against faster photoevaporation — predicting the Horsehead survives $\sim 5\%$ longer than SM PDR models estimate (i.e., $1.05\times$ SM lifetime).

Q5 Prediction 3: $B = 10^{-6}$ T (weaker than most molecular clouds in the per-system series) predicts that the Horsehead Nebula has a mass-to-magnetic flux ratio $M/\Phi_B = M/(Br^2) = 1.989 \times 10^{33}/(10^{-6} \times 5.59 \times 10^{32}) \approx 3.56$ (supercritical) — meaning magnetic support is insufficient to prevent collapse and the pillar is gravitationally unstable on ~ 1 Myr timescales at its tip. Testable via JCMT SCUBA-2 polarimetric maps of dust emission polarization.